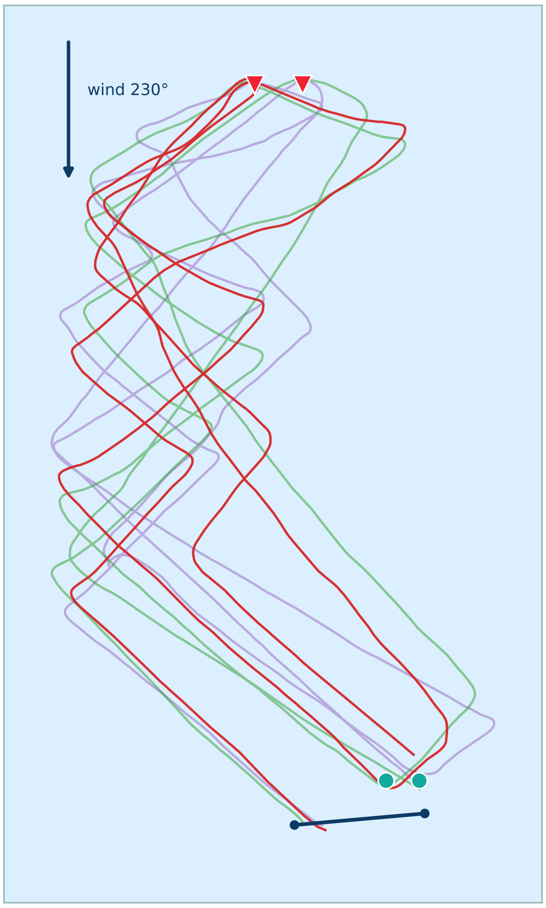
TELEMETRY PERFORMANCE REPORT

MOTH — Race 8

Lago di Garda, Malcesine · 5 Jul 2026 · Start 14:01 (local)

Race course — all boats



Fleet

- CRO 5087
- AUS 5250 *
- ESP 56

Race winner

- CRO 5087 finished first in this race.
- 2nd: AUS 5250 * 3rd: ESP 56
- Afternoon breeze; W-L-W-L course.
- Wind, marks and course axis estimated from GPS (no wind sensor in logs).

Instrument note (*)

- AUS 5250: speed sensor mounted rotated. Heel & heel-stability are rebuilt from the boat's gravity + course — an accurate estimate, not exact to the degree.

Positions at the marks — gains (green) / losses (red)

Finish	Boat	M1	M2	M3	Fin	Δ M1→M2	Δ M2→M3	Δ M3→Fin	Net
1	**CRO 5087**	1	1	1	1	+0	+0	+0	+0
2	**AUS 5250 ***	2	2	2	2	+0	+0	+0	+0
3	**ESP 56**	3	3	3	3	+0	+0	+0	+0

M1 = windward 1 · M2 = leeward 1 · M3 = windward 2 · Fin = finish. Positive Δ (green) = places gained on that leg.

Start analysis — 14:01 gun

Fin	Boat	Line pos	Bias loss (m)	DTL gun	DTL -10s	SOG gun	SOG -10s	TWA gun	SOG 1′	Heel stab 1′	VMG 1′
1	**CRO 5087**	3-1	24	17	37	23.3	21.6	73	19.0	3.8	12.7
2	**AUS 5250 ***	3-3	28	3	17	21.2	10.0	60	19.4	5.7	13.5
3	**ESP 56**	3-2	24	6	21	22.9	20.6	78	18.2	4.7	11.4

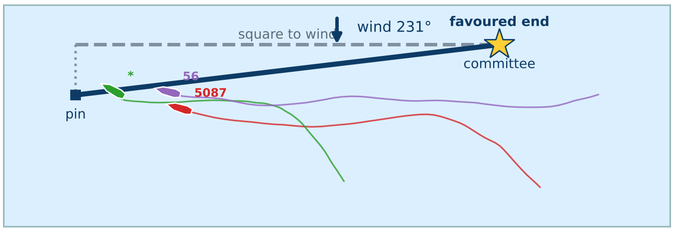
How to read the start

- Line pos: the line is split into 3 thirds, each third into 3 — read third-subpart from the committee end (1-1) to the pin end (3-3).
- Bias loss (m): metres given away at the gun from line position alone vs the favoured (pin) end, from the TWD bias.
- DTL = distance to line (m, negative = over early). 1′ = first minute after the gun; heel stability = avg heel change per 10s (lower = steadier).

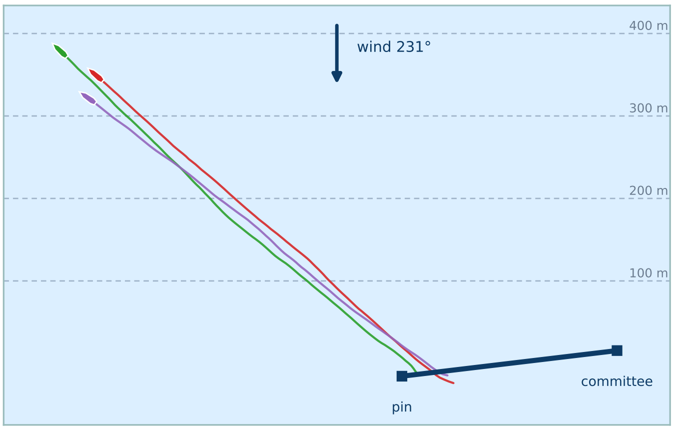
Line bias at the start (vs TWD)

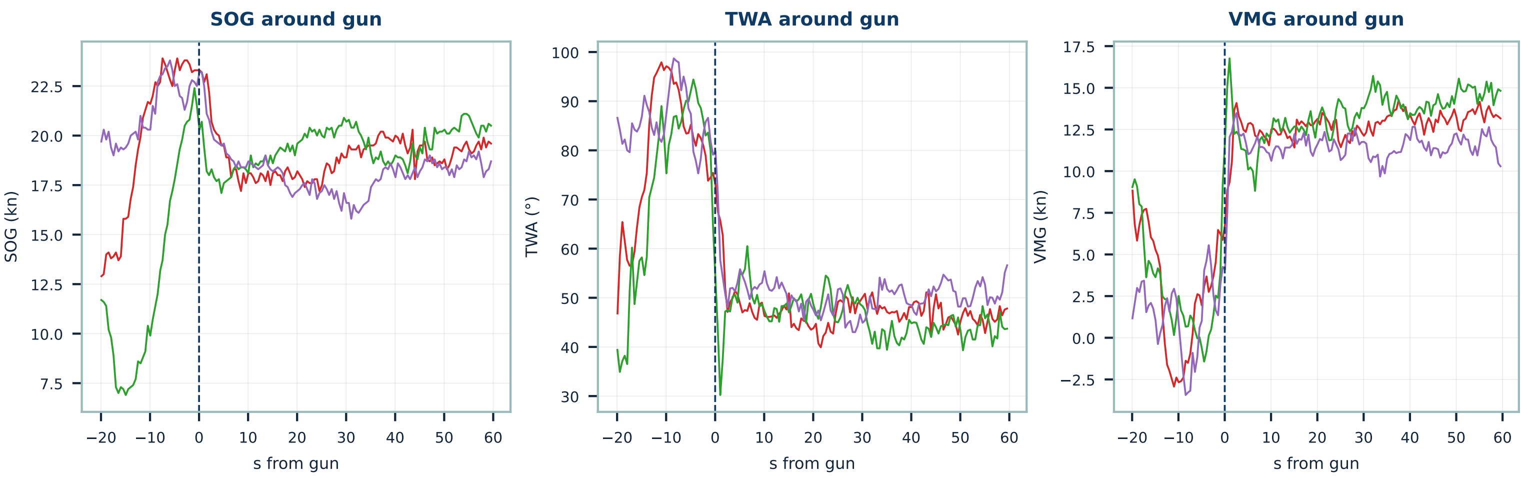
- Start line 263 m, bearing 314°.
- Wind at start ≈ 231°, estimated from the first beat.
- Line bias vs TWD: 6.8° — Committee end favoured.
- Advantage ≈ 31 m (≈ 6.2 boat-lengths) for a boat starting at the committee end.

Position at gun

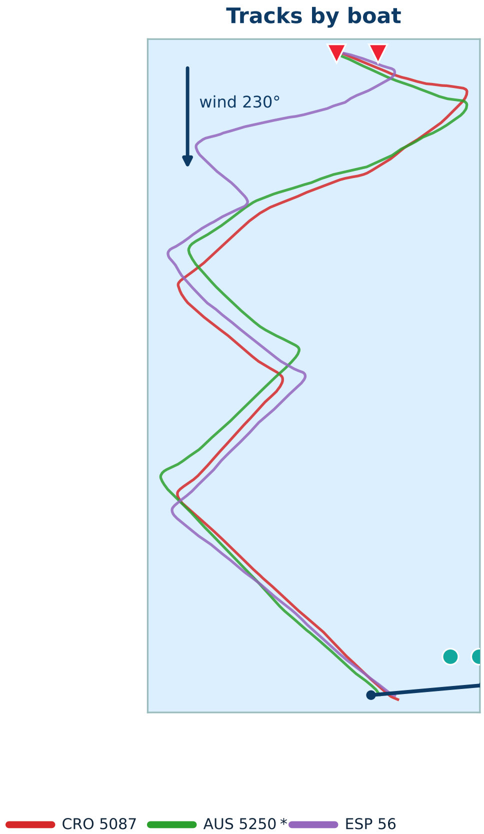


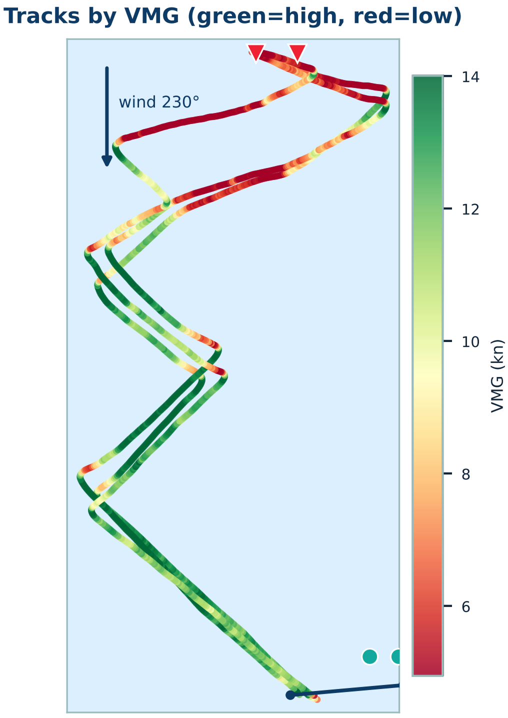
Position 1 min after gun





Leg 1 · Upwind

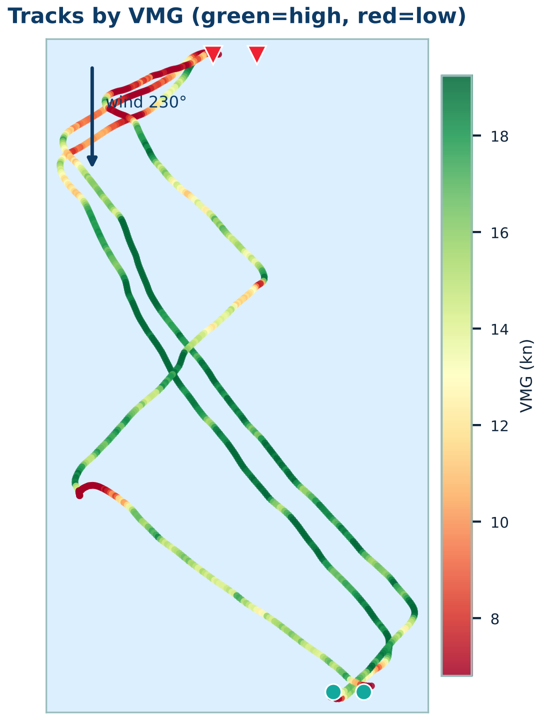




Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Tacks
1	**CRO 5087**	10.40	53	17.58	20.30	21.3	9.5	-0.5	+0.1	2535	281	4
2	**AUS 5250 ***	10.14	55	17.63	20.70	21.9	11.9	-3.7	-1.3	2595	286	4
3	**ESP 56**	9.70	54	16.86	19.90	22.1	15.1	-4.3	-0.2	2588	299	6

Rows ordered by Avg VMG (best first); the Finish column keeps each boat's race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / − more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.

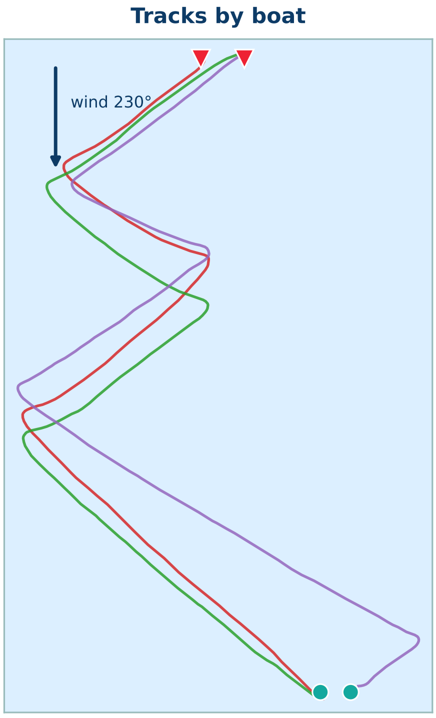
Leg 2 · Downwind

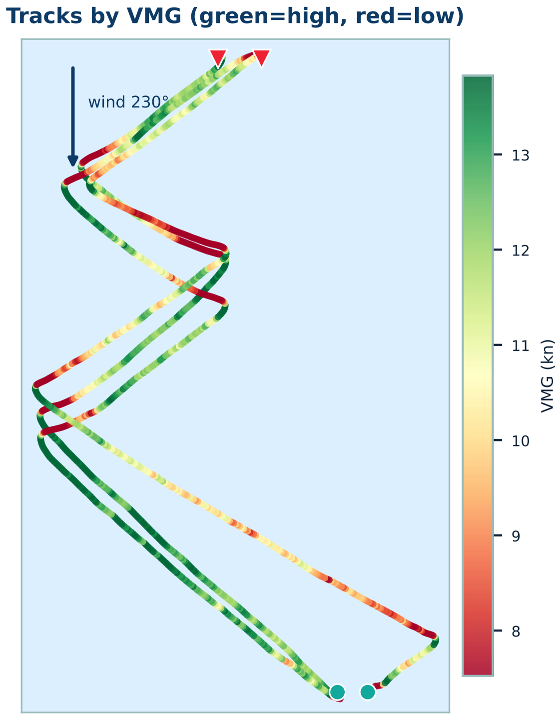


Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Gybes
1	**CRO 5087**	15.65	140	21.17	25.40	11.2	6.6	+0.9	-0.2	1906	174	2
2	**AUS 5250***	15.50	137	21.50	25.50	12.2	6.2	-1.4	-0.8	1956	177	2
3	**ESP 56**	12.16	126	19.65	26.40	12.1	15.7	-0.1	-0.2	2270	238	3

Rows ordered by Avg VMG (best first); the Finish column keeps each boat's race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / − more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.

Leg 3 · Upwind





CRO 5087 AUS 5250* ESP 56

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Tacks
2	**AUS 5250***	11.60	53	19.46	21.30	23.5	13.1	-2.3	-1.7	2377	237	3
1	**CRO 5087**	11.43	50	18.32	20.50	20.8	11.6	-0.4	+0.1	2280	242	4
3	**ESP 56**	10.10	57	18.65	21.80	22.7	10.3	-2.7	+0.1	2563	267	4

Rows ordered by Avg VMG (best first); the Finish column keeps each boat's race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / − more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.

Leg 4 · Downwind

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Gybes
2	**AUS 5250***	16.91	136	24.61	28.04	14.9	9.1	-0.8	-1.0	2072	164	2
1	**CRO 5087**	16.04	137	22.28	25.20	13.8	8.9	+1.4	+0.0	1886	164	3
3	**ESP 56**	15.47	137	21.59	24.19	9.4	7.3	+2.3	-0.2	1971	177	2

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Tacks analysis

Finish	Boat	# Tacks	Entry	Exit	Min	SOG Δ	Angle Δ	Turn °/s	Heel in	Heel out	Dist loss (m)	VMG loss (m)
3	**ESP 56**	10	18.7	14.0	12.0	6.7	+5	118	20	21	26.1	22.8
2	**AUS 5250 ***	7	19.0	14.3	12.4	6.7	+14	117	18	21	26.2	35.5
1	**CRO 5087**	8	18.6	13.6	11.6	7.0	+13	118	17	22	26.8	33.9

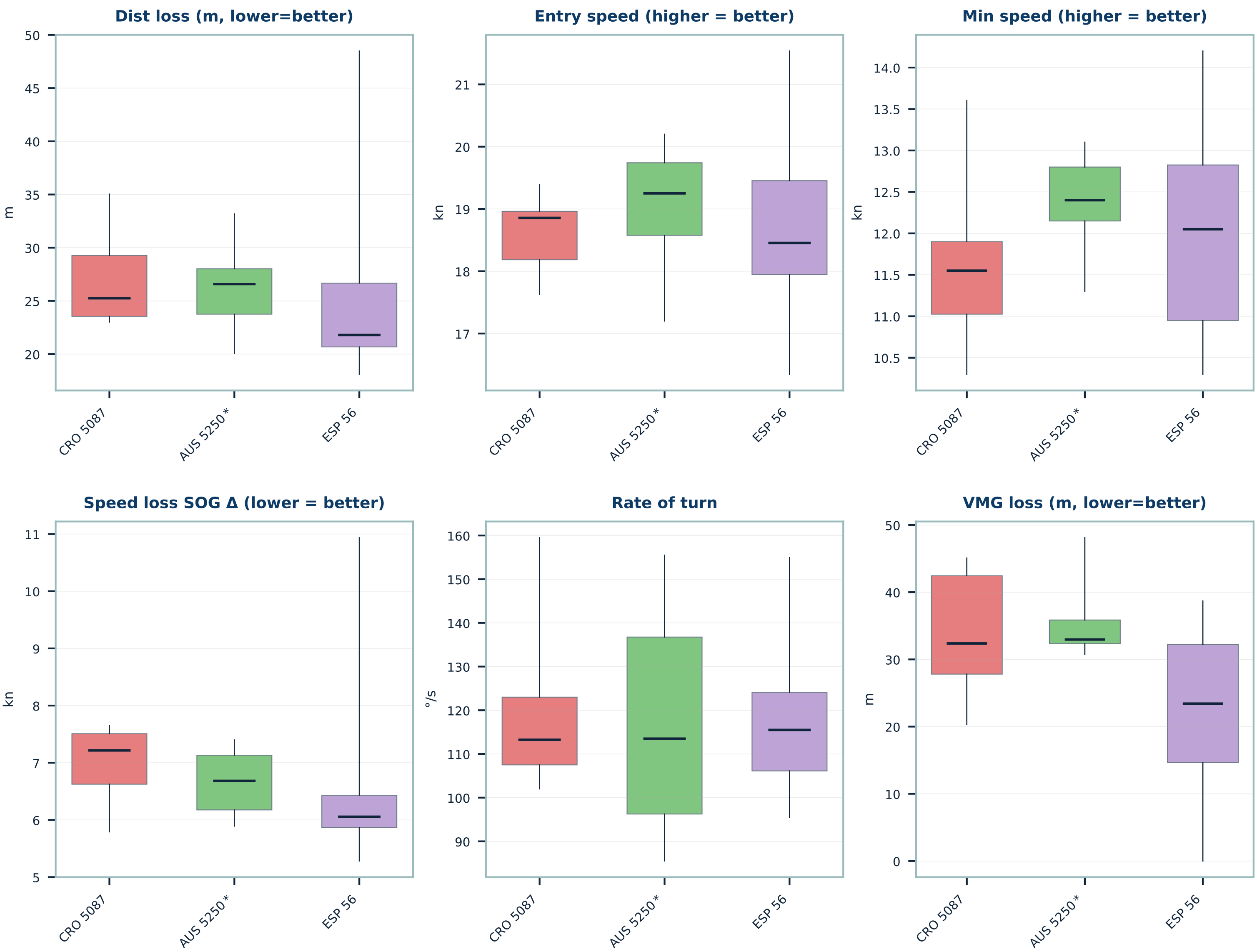
Rows ordered by Dist loss (most efficient first); the Finish column keeps each boat’s race finish position.

Tacks analysis — metric definitions

- Entry / Exit SOG = mean speed over 6 s before / 9 s after the manoeuvre. Min SOG = lowest speed in between.
- SOG Δ = Entry − Min (how much speed was bled off). Angle Δ = exit TWA − entry TWA (+ = exits wider).
- Rate of turn = peak heading change (°/s). Heel in/out = mean |heel|.
- VMG loss = made-good (toward the mark) deficit vs holding entry VMG. Dist loss = actual distance lost from the speed drop vs holding entry speed.

How to read the charts below

- Each box = one boat, ordered left→right by race finish. The plots span all of that boat’s manoeuvres.
- Whiskers = min & max; box = middle 50% (25th–75th pct); line in the box = median. A lower, tighter box = more consistent.
- VMG/Dist loss, Speed loss, Min-speed: lower is better. Entry & Min speed: higher is better.



Gybes analysis

Finish	Boat	# Gybes	Entry	Exit	Min	SOG Δ	Angle Δ	Turn °/s	Heel in	Heel out	Dist loss (m)	VMG loss (m)
2	**AUS 5250 ***	4	22.1	19.3	17.3	4.8	+4	86	14	9	15.5	9.5
1	**CRO 5087**	5	21.7	18.0	16.8	4.9	-4	81	11	9	21.0	21.9
3	**ESP 56**	5	20.3	15.2	12.5	7.8	+6	237	13	20	28.4	17.7

Rows ordered by Dist loss (most efficient first); the Finish column keeps each boat's race finish position.

Gybes analysis — metric definitions

- Entry / Exit SOG = mean speed over 6 s before / 9 s after the manoeuvre. Min SOG = lowest speed in between.
- SOG Δ = Entry − Min (how much speed was bled off). Angle Δ = exit TWA − entry TWA (+ = exits wider).
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